COURSE OUTLINE

* Survey design, planning and Programming
* Methods of data Collection
* Errors and Biases
* Estimation of Mean, totals, ratio and Proportions in Simple random Sampling
* Systematic random Sampling
* Stratified random Sample



Survey Design

A survey design is the process of planning, developing and implementing a survey to ensure the data collected is valid, reliable, unbiased and answers a research objectives with minimum errors and cost.

Survey? A survey is a systematic method of gathering information from a sample (a sample of) entities using a standardized set of questions for the purpose of making inferences about the whole population from the samples studied.

TYPES OF SURVEY

1. Cross-Sectional Survey:- In Cross-Sectional Surveys, data collected at one point in time is selected to represent a larger population, the researcher simultaneously accesses the participants exposure and outcomes.

2. The Mixed Survey Design This uses more than one approach (Method) in the same survey to collect better data. It could be quantitative and qualitative.

3. Panel Survey Design:- A panel survey design is a long-term research that tracks people's behaviours over time including their thoughts, feelings, beliefs and

erations. The survey is repeated overtime and it involves same people.

4. Longitudinal Survey Design: It is an Observational one just as Cross-Sectional one, when researchers observe the same participants repeatedly for several years. Longitudinal survey measures the same variables at different time point to see how things change, grow, decline or stay stable.

5. Prospective survey design: It examines trends, forecasts and results by starting in the present and following participants into the future.

6. Cohort Survey design: It is also a longitudinal study, it involves keeping track of research subject or groups across time.

7. Qualitative survey design: It makes use of open-ended questions to generate lengthy narrative or typed responses in order to extract views, points, experiences, narratives etc.

It is using a survey to collect a non-numerical data instead of just counts.

ESSENTIAL TERMS IN SURVEY DESIGN

* Target population: The whole group about

when you wish to make conclu
sions
* Response Rate. The percent
age of respondents that
completed the Survey from
the Sample is known as
Survey response rate - it is
the proportion of respond
ent to the poll out of all
those asked to participate.
* Demographics: This means
characteristics of a popn that
describes who people are.
it is also statistical data
about the structure of the
popn, age, sex, education
income, marital status
etc.
So a demographic Survey
gathers data on the grou
ps fundamental
traits.
* Survey Sampling: This is

the process of sele
smaller survey
large popn. In
to study the small
and make conclusion
about the larger popn
* Survey Bias: This
- any systematic err
that makes your re
result consistently
wrong.
* Reliability: The de
to which a survey
ces a stable and cons
result with low rand
errors.
* Validity: It is about
accuracy of the measu
more. It is the degr
to which a survey
collects data it suppose
to contain. It is the deg
to which a survey meas

as the concept it is supposed to measure.

STEPS IN PLANNING AND EXECUTION OF A SURVEY

1) Objective of the study: It is necessary to state the objective of the study in precise, clear and concise terms. It is also needful to state the purpose for which the objective will be used. This will help to avoid being distracted in decision making in the course of the study.

2) Target Population: It has to be clearly defined. The pop can be defined in terms of the content, unit extent and time. E.g everyone with a certain age range [Content], primary and secondary school students [Units], people living in a particular location [Extent], a certain time or period under consideration [Time]. The pop to be studied is to be the same with the pop about which information is needed.

3) Data to be Collected: A researchers must ensure that all data to be collected are relevant and essential to the purpose of the survey.

4) Method of data Collection: Data Collection can be either with the use of Questionnaire, Personal interviews, physical

Observations. Laboratory Measurements, Field Investigations, Documented data etc.

5. Sampling-frame and Units
A list of the item or people to be covered during the sampling process, together with identifiable schemes for selecting them. It is also a list of Sampling Units. Sampling Units are finite members of distinct and identifiable Units. During Sample the pop must be capable of division into Sampling Unit for Sample Selection. It must Cover the Whole pop.

6. Degree of precision required
It is the extent of how precisely an Object or parameters of the study is measured. It important to always specify the degree of precision needed for the various parameters to be estimated.

- Data collection methods
Interview Method: interview involves direct verbal communication between a researcher and a respondent allowing questions to be asked and answered. In real term interview can be carried out face to face, Over the phone or through Online platforms such as Zoom or Sky.

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Sampling Survey and Survey Method

Design of forms and Questionnaire

In survey research, the quality of data collected depends greatly on the quality of the questionnaire or form used for data collection.

A questionnaire is a carefully designed instrument containing a set of questions used to obtain information from respondents.

A form refers to the structured layout used for recording responses, especially in official surveys and administrative data collection.

The design of forms and questionnaire is these for a critical stage in sampling surveys because poorly designed instruments can lead to inaccurate responses, bias, non-response, misunderstanding and unreliable conclusions.

A well-designed questionnaire should:

- Collect relevant information accurately.
- Be easy for respondents to understand.
- Minimize errors and bias.
- Encourage cooperation and participation.
- Facilitate easy coding analysis and interpretation.

MEANING OF QUESTIONNAIRE DESIGN

Questionnaire design is

1) the systematic process of cons
structing questions and assa
ranging them in a logical order
2) to obtain reliable and valid
information from respondents
3. It involves:
1) Identifying Survey Objectives
2) Determining the information
needed
3) Selecting question types
4) Wording questions properly
5) Organizing questions
logically
6) Designing response format
7) Pre-testing the Instrument

OBJECTIVES OF QUESTIONNAIRE DESIGN

The major objectives of the
questionnaire design include:
6. 1) To obtain accurate and
relevant data
2) To reduce measurement
errors.

8) To encourage
to provide
answers.
4) Brevity:
should be
Concise
5) Neutrality
should not imp
respondent.
Biased "Do yo
Online learning
than classroom
Neutral: "Which
Learning pro
6) Logical
Questions should
natural flow:
E.g General ques
ould come before
One's
7) Ease of Tabulat
Responses should
code and analyze

cally:

Types of Questionnaires

i) Structured Questionnaire:
This contains fixed questions with pre-determined response options.
This kind of questionnaire can be easy to analyze, its responses are Uniform and its suitable for large Surveys. E.g Yes/No, True/False

Advantages
* Detailed Informations
* Useful exploratory Studies

Disadvantages
i Difficult Coding
ii Time-Consuming analysis

ii) Close-ended Questions:
Here, respondents select from provided alternatives.
E.g: "Do you own a Smartphone
- Yes
- No

Advantages
a) Easy Analysis
b) Faster Responses

Disadvantages
* Limited Information
iii) Dichotomous Question
Here, Only two response Options is provided

iv) Multiple Choice Question
Here, Several response Options are provided
e.g "What is your highest education Qualification
- Primary
- Secondary
- Diploma
- Degree
- Post-Graduate

v) Rating Scale Questions:

Continuation of Note

The flexibility of ... Suitable for ... of Studies especially ... that examines deeper ... hts into people's ... attitudes or behavior ... The key strength of ... lies in the richness ... information they ... They not only capture ... verbal responses ... allow the researcher ... observe tone, expression ... and other non-verbal ... gestion. However the ... are often more costly ... and time consuming

Likert scale questions are used to measure attitude & opinions. Could be called the "Likert Scale Question".

Eg. Rate your satisfaction with Online Classes?

Response	Score
Strongly Agree	5
Agree	4
Neutral	3
Disagree	2
Strongly Disagree	1

iv) Ranking Questions: Here, respondents rank items according to preference.

E.g. Rank the following modes of Transportation.

2.) Questionnaire Method

A questionnaire is a structure set of written questions used to collect information from

respondents. It can be designed to obtain numerical data such as income or age and or qualitative insights such as opinions and attitudes Unlike interviews, questionnaires are usually self-administered which makes them a cost effective way of reaching many people. Questionnaires may be administered in different ways including paper forms, online platforms such as google forms mobile based surveys or emails.

There are two main types of questionnaires

- Open-Ended Questionnaire: It gives respondent freedom to express their views in their own words which provides richer details but can be hard to analyze.

- closed-Ended Questionnaires: It restricts answers to pre-set options making, responses easier to compare and analyze in practice. Many studies combine both types for balance.

3.) Observation Method

Observation is a systematic method of data collection in which the researcher directly observes and records the behaviours, events or conditions as they occur either in natural environment or under controlled settings. Unlike questionnaires or interviews that relies on self reports, observation

captures actual behaviours as it happens revealing data that might not surface through self-report such as body language, spontaneous reaction, group dynamics.

However, it can be time consuming limited by ethical concerns and influenced by researcher's bias.

Types of OBSERVATION

*participant Vs Non-participants: In participant Observation, the researcher actively engage with the group while observing whereas in Non participant Observation, the researcher remains detached only watching and recording

*Control Vs Naturalistic-Control Observation: Takes place under conditions arranged by the researches such as laboratory experiments while Naturalistic Observation occurs in real world setting without interference

*Structured Vs Non-Structured: Structured observation uses checklists or schedules to record predefined behaviours while Unstructured Observation is more flexible allowing the researchers to note events freely as they unfold.

* Focus Group: A focus group is a method of collecting data by bringing together a small group of participants (6-12) to discuss specific topics under the guidance of in-

cially useful for exploring perceptions, experiences and opinions.

* Mixed Methods of Data Collection: Mixed Method involves Combining two or more data collection approaches within a single study such as using survey for broad patterns and interviews and observations for deeper insights.

BASIC CONCEPTS OF SAMPLING

1.) DEFINE SAMPLING AND EXPLAIN

ADVANTAGES OF SAMPLING AND ITS DISADVANTAGES

DEFINITION OF TERMS

1. Elementary Units or Unit: This is an element or group of elements from which the required information is obtained e.g household, farm, persons etc.

2. Reporting Unit: This is the unit that usually supplies the needed information. E.g The head of the household giving information about the household.

3. population: This is the collection of unit having some features in common. it can also be refered to the entire group of individuals items or event of interest.

4. Sampling Units: Those are elementary unit of a population from which a sample is selected.

5. Sampling Scheme: This is a method of selecting a sample from a population.
E.g Simple random sampling
Systematic random sampling
Stratified random sampling
etc.

6. Sampling Design: Is the collection of all possible design together with their probabilities of selection

7. Sampling Strategy: Is the pair of the sampling design and estimator.

eg. Precision: This is a measure of how close an estimate is to its average value over all possible sample.

Study Errors Vary from one form of sample to the other but are usually minimized when the number of sample increases

These errors can occur due to the following:

a. Faculty Selection of the Sample.

b. Faculty Substitution

c. Faculty demarcation of Sampling Units.

Sampling errors are measured by Sample variances.

DATA COLLECTION ERRORS

Sampling Error: This is errors due to the Method of Sampling used in the

Non-Sampling Errors: These are errors that occur at the stages of data collection Compilation or Computation i.e Analysis

These errors are further classified into response errors and Non-response error.

=> Response Errors: Are errors from the interviewer instrument of data collection such as questionnaire and the respondent.

=> Non-response Errors: Can be inform of non-Coverage item, non-response and total Unit response.

"= Errors: These are systematic errors, this error increase with the Sample Size. The following may be the Cause of bias error:

i Deliberate:- Selection of respondent or items.

ii Substitution of One respondent or item directly chosen in the sample with another.

iii Feature to Cover the whole Selected Sample.

iv Haphazard Selection.

STEPS IN DESIGNING A QUESTIONNAIRE

1.) Define the Objectives of the Survey. Here the researcher must clearly identify; What information is needed, Why it is needed and How it will be used.

2.) Identify the target Respondents: Here we determine who will answer the questionnaire, the educational level, Language ability and Social Background

3.) Choose the Method of Data Collection: Questionnaire design depends on whether the Survey is face to face interview, telephone Survey, Online Survey or Mailed Questionnaire

4.) Determine the type of Questions: Here, we choose the suitable question Format such as; Open ended, Close-ended, Rating Scales, or Multiple choice

5.) Draft the Questions: While drafting, we use Simple Language, No-ed technical terms and double Barrelled Question:

E.g of Double Barrelled Question is "Do you enjoy Statistics and Mathematics?" - This is a double barrelled question because respondent may enjoy but not the Other.

6.) Arranged Questions properly: The recommended Order for a Question design is given below

i Introductory Question
ii Simple Questions
iii Close Survey Questions
iv Sensitive Questions
v. Demographic Questions

7.) Design the questionnaire layout: A good layout should include
a. Title
b. Instructions
c. Numbering
d. Adequate Spacing
e. clear response area

8.) Pretext Or Pilot Test: The questionnaire should be tested On a small Sam ple before the main Survey. This is important because it helps to detect ambiguous questions, identify Missing Options Measures Cooperation time and it improves reliability.

LAYOUT

* Title
Clearly state the Survey Topic

* Introduction
Explain the Purpose of Study, Confidentiality assurance and importance of participation

* Instruction
Instruction should be simple and visible

* Numbering
Questions should be numbered properly

* Spacing:
Adequate Space should be provided for responses

* Coding Space:
Space may be reserved

for coding and data entry.

Some SOURCES of ERROR IN QUESTIONNAIRE DESIGNS

Below are Some of the Sources of essor encounte red in Questionnaire design:

- Response Essor: The respondent may give inacc- urate answers.
- Interviewer Bias: In this case, the interviewer may have influenced the answers or responses of the respon dent.
- Question Wording Essors: Here, the essor may be due to poor wording causing misunderstanding
- Non-Response Essor: This is a case where the respon dent fail to answer some questions.

ADVANTAGES OF WELL DESIGNED QUESTIONNAIRES

- It yields accurate collection.
- It reduces bias
- It gives room for access.
- It promotes a high response rate.
- It promotes better decision Making.

LIMITATIONS OF QUESTIONNAIRES

1) Some respondents may misunderstand the ques tion.
2) Low response rate may occur
3) Illiterate respond may face difficulties
4) Open ended resp nses may be difficult to analyze.

DATA PROCESSING ANALYSIS AND INTERPRETATION IN SAMPLING SURVEY AND SURVEY METHOD

In Sampling Survey and Survey Methods, data are collected from a sample of a popn in other to make valid Conclusions about the entire popn.

After data Collection, the next important stage are

i) Data Processing
2) Data Analysis
3) Data Interpretation

This Stages helps the researcher Convert raw data into meaningful information for decision making and policy formulation.

DATA PROCESSING: Refers to the systematic procedure of converting raw Survey data into a form suitable for analysis and interpretation. It involves Organizing, checking Coding, Classifying and summarizing Collected data.

OBJECTIVES OF DATA PROCESSING

a. To improve data quality.
b. To detect and Correct errors
c. To Organize data for analysis
d. To ensure reliability and accuracy
e. To Simplify Computation and interpretation

STAGES IN DATA PROCESSING

1. Editing: Is the examination of Collected questionnaires or Schedules to detect and Correct errors and Omission

2. TYPES OF EDITING

We have two types of editing

* Feed Editing
* Central "

FIELD EDITING: This is an editing done immediately data is Collected by the interviewers.

CENTRAL EDITING: This is an editing done by the Supervisors or Statisticians

Below are the Common errors Corrected during editing

i) Missing Value
ii) Inconsistent Response
iii) Illegible Entries
iv) Wrong Unit of Measurement

CODING: Involves assigning numerical Symbols to responses for easy analysis.

Example:

Response	Code
Male	1
female	2

ADVANTAGES

A. Simplifies Analysis
B. Saves Time
C. facilitates Computer entries.

3. CLASSIFICATION: This is the grouping of data into homogenous Categories

Types of Classification

1. Qualitative Classification
E.g Gender, Religion
2. Quantitative Classification
E.g Age Groups, Income level
3. Geographical Classification
E.g States, Regions etc.
4. Chronological Classification:
E.g Years, Months.

Age Group	
10-19	12
20-29	18
30-39	10

4. TABULATION: This is the presentation of data in rows and columns.
Below are the types of Table:
i Simple Table
ii Double vv
iii Complex vv

Assignment

DATA ANALYSIS: This is the process of examining and summarizing data in Orders to draw conclusions. In solving planning, Analysis usually involves estimation of popn characteristics for Sample data.
In popn Mean, popn total, popn proportion and the ratio estimation are the measures Commonly estimated

Survey Sampling

Estimation in Survey Sampling

Estimation is the use of sample statistics to infer the value of unknown population parameters

Types of Estimation

1. Point Estimation
2. Interval ''

The population mean is the average value of all that is in the population

The following notations are used in the estimation of population mean

i. Population Size
ii. Sample Size
iii. Observations: $y_1, y_2, \ldots, y_n$

The sample mean is used to estimate the population mean

ESTIMATOR OF POPULATION MEAN

$\bar{y}$ = Sample mean = $\frac{1}{n}$

Where; y = Sample

n = Sample size

Under simple random sampling without replacement (SRSWOR)

$$\hat{\bar{Y}} = \bar{y}$$

(estimate of the population mean)

Variance of Sample ($\bar{y}$)

$$Var(\bar{y}) = \left(1 - \frac{n}{N}\right)\frac{S^2}{n}$$

Where S^2 = population variance $= \frac{\sum_{i=1}^{N}(y_i - M)^2}{N}$

And

$\left(1 - \frac{n}{N}\right)$ is the finite population correction factor

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Assignment

* Simple Table: Is the arrangement of data according to Only One characteristic Or Variable. It presents information in a single Way for easy Understanding.

* Double Tabulation: Is the arrangement of data according to two characteristics Or Variables at the same time. It helps in Comparing two related sets of information.

* Complex Tabulation: Is the arrangement of data using more than two characteristics Or Variables. It provides detailed information and allows deeper analysis.

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Assignment

1. Define Sampling and Explain

Sampling is the process of selecting a smaller group of people, items Or Observations from a larger population for the purpose of study Or research.

Explanation

Sampling is a method used in research to choose a small number of people Or items from a large group in Order to study them and gather information. The large group being studied is called the population, while the smaller group selected from it is called the sample.

Sampling helps researchers because studying the whole population may be too successful, expensive or take too much time. Instead researchers study only a part of the population and use the findings to represent the entire group.

Advantages of Sampling

* Saves Time.
* Reduces Cost
* Easy to Manage
* Provides quick Result
* Useful for Large Populations

Disadvantages of Sampling

* Risk of Bias
* less accurate than Census
* Sampling errors Can occur
* Requires Skill

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TYPES of SAMPLING

1) Non-Probability Sampling (Random Sampling)

In Non-Probability Sampling, it is the probability of selecting a unit in the sample is not known and cannot be determined in the Sampling.

In other words, a unit in the sample is not Statistical inferences Cannot be made Objectively about the population a non-random Sampling.

Example

* Judgemental
* Quota

1-) Judgemental Sampl

In the sampling, expert pick that they judge to be represe ntative of the popn. E.g a typical town may be picked to represent an Urban popn a village picked to represent a rural popn. Judgement sampling is used whenever such limitation are extreme hetergenity of the popn, time and cause cost it necessary to choose small sample.

2.) Quota Sample. Specifies no of Sample unit in each specified group, based on say, Gender, age, Social class etc. Is assigned to an enumerators.

Quota Sampling can therefore to be likened to Stratified Sample in each ~~pattern~~ Stratum is not Selected by probability Sample.

Quota Sampling is used extensively in opinion Survey and in market research.

ADVANTAGES of Quota Sampling

1.) Its easy to apply and avoid the problem of not at homes and callbacks

2.) Its useful in obtaining accurately the people's reaction to the current issues.

3.) Quota Sampling can be used even where there is no suitable list of population limit.

DISADVANTAGE

* It may not be possible for the interviewers to fill the quota with representative unit.

* Sampling Variability cannot be calculated with Quota Sampling.

PROBABILITY SAMPLING OR RANDOM SAMPLING

Probability Sampling is a Sample of element drawn from the popn of element in such a way that every element in the popn has a known and Non-Zero probability of being selected.

Examples:

i Simple random Sampling [SRS]

ii Stratified random Sampling.

iii Systematic random Sampling

v Cluster Sampling

ot ii Simple random S
ling: A Sample is co
Simple random Sam
ling of each member
of the popn has a kn
and equal probability
of being included or
selected in the Samp
In other words a Sim
random Sampling
is a random Sa
mple in which membe
of the popn has eq
ual probability of
being drawn in the
Sample.
This method is free
from bias.
One method of obtai
ning a random sample
is to numbers or name
every item or unit in
the popn and Mix in

them up thoroughly, items are drawn one after the other using lottery method until the required sample size are obtained. Another method is through the use of Table of random numbers

Advantages of Simple Random Sampling

1) provides us with reliable estimates of the unknown popn we wish to study
2) It enables us to make value estimate of the sampling error.
3) It's easy to carry out calculation
4) It's straight forward.

Disadvantages

1. Limited to homogeneous popn.
2. Can be ineffective if not properly handled.
3. Expensive when the popn is large.
4. Under Simple random Sample we have
   * Sample with replacement.
   * Sample without replacement

Sampling in which the elementary unit of the popn is more than one is called Sampling with replacement

Formula:

N^n

Sampling in which the unit of the popn is not available for selection is called Sampling

Where sep/warrent formula:

N_{Cn}

Most sample surveys deals with sample drawn from population of finite no of units N. If the (N) units are all distinguishable from one another, the number of distinct Samples of size (n) that can be drawn without replacement from (N) unit is given by:

N_{Cn}

If the selection is with replacement, the no of distinct sample of size (n) that can be drawn is given by

N^n.

In selecting a simple random sample of small n element without replacement from a population of N element, the following should be noted

1. The first sample element is selected by giving each of the N population element equal probability $\left[\frac{1}{N}\right]$ of being selected.
2. The second sample element is selected by giving each of the remaining $(N-1)$ population element equal probability $\left[\frac{1}{N-1}\right]$ of being selected.
3. The process is repeated until all the small sample elements have being selected.

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Some Useful Symbols and Formulae

Basic	Population	Sample
Total	$T = \sum_{i=1}^{N} y_i = y_1 + y_2 + \dots + y_N$	$T = \sum_{i=1}^{n} y_i = y_1 + y_2 + \dots + y_n$
Mean	$\bar{Y} = \frac{\sum_{i=1}^{N} y_i}{N}$	$\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n}$
Variance	$S^2 = \frac{1}{N-1}\sum_{i=1}^{N}(y_i - \bar{Y})^2$ $= \frac{1}{N-1}\left(\sum_{i=1}^{N} y_i^2 - N\bar{Y}^2\right)$	$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2$ $= \frac{1}{n-1}\left(\sum_{i=1}^{n} y_i^2 - n\bar{y}^2\right)$

Mean $\bar{Y}$.

Total Y

Ratio of the totals = $R = Y/X$

ratio of the two means $R = \frac{\bar{Y}}{X}$

Note:

(a) $\bar{y}$ is used to estimate $\bar{Y}$

(b) $\hat{Y} = N\bar{y} = N\left(\frac{\sum_{i=1}^{n} y_i}{n}\right) = \frac{N}{n}\left(\sum_{i=1}^{n} y_i\right)$ is used to estimate the population total (Y)

(c) $\hat{R} = \frac{\bar{y}}{\bar{x}} = \frac{\sum y_i}{\sum x_i}$ is used to estimate R i.e population Ratio.

(d) in $\hat{Y} = \frac{N}{n}\left(\sum_{i=1}^{n} y_i\right)$, the expression $\frac{N}{n}$ is called expansion raising inflection factor. It's the inverse of the sampling fraction. $f = \frac{n}{N}$!

$\bar{y}$ is an unbiased estimator of the popn mean and its variance is given as

$$Var(\bar{y}) = \frac{\sigma^2}{n} = \left(\frac{N-1}{N}\right)\frac{S^2}{n}$$

for Simple random Sampling with replacement.

$$Var(\bar{y}) = \left(\frac{N-n}{N-1}\right)\frac{\sigma^2}{n}$$

$$= (1-f)\frac{S^2}{n}$$ for Simple